

Project Description

Lightweight LLM Fine-Tuning for Query Expansion on MS MARCO

Deadline: August 21, 2026, at 11:59 PM

1. Overview and Objectives

The goal of this assignment is to design, implement, and evaluate a lightweight open-weight large language model (LLM) for query expansion in sparse (BM25) retrieval, following the [Query2Doc](#) paper. You will compare retrieval effectiveness across three configurations: **(1) zero-shot generation**, **(2) supervised fine-tuning (SFT)**, and **(3) SFT followed by Direct Preference Optimization (DPO)** with a retrieval-based preference signal. The assignment prioritizes methodological soundness, reproducibility, implementation quality, and depth of analysis over absolute retrieval effectiveness.

2. Models, Data, and Tooling

- **Model:** choose one lightweight open-weight LLM that can be fine-tuned on a laptop GPU. Suggested starting points are Qwen/Qwen2.5-0.5B-Instruct, Qwen/Qwen2.5-1.5B-Instruct, or google/gemma-3-1b-it. Other lightweight LLMs are acceptable if you justify the choice against the compute budget.
- **Fine-tuning data:** [MS MARCO Passage \(V1\)](#) corpus, MS MARCO training queries (~500K), and the associated relevance judgments (qrels).
- **Evaluation data:** TREC Deep Learning 2019 (passage task) queries and qrels.
- **Tooling:** [Pyserini](#) with a prebuilt BM25 index for retrieval

3. Query Expansion Prompt

Use the following prompt verbatim to generate a pseudo-document for each query:

```
"""
```

```
Write a passage that answers the given query:
```

```
Query: {query_text}
```

```
Passage:
```

```
"""
```

Cap each generated passage at **128 output tokens**. Remaining decoding parameters (temperature, top-p, etc.) are your decision and must be justified. For reproducibility, evaluation-time generation should be deterministic or seed-fixed; note that Stage 3 requires sampling to obtain diverse candidates.

4. Task Specification

Stage 1 — Zero-shot baseline

Without any fine-tuning, generate one pseudo-document per query. Construct the expanded query following Query2Doc: concatenate the original query **repeated five times** with the generated

pseudo-document (the repetition up-weights the original query terms under BM25). Retrieve with BM25 in Pyserini and evaluate on TREC DL 2019 queries.

Stage 2 — Supervised fine-tuning (SFT)

Fine-tune the selected model on MS MARCO so that it maps a query to a relevance-judged passage for that query. Wrap the training input in the same prompt template used at inference, so that training and generation are consistent. **The number of training examples is your decision** and must be justified. Evaluate with the identical retrieval and scoring pipeline.

Stage 3 — Direct Preference Optimization with retrieval feedback

Starting from the SFT checkpoint, for each training query (drawn from a set disjoint from both the SFT queries and the evaluation queries):

- sample multiple candidate pseudo-documents;
- run BM25 retrieval for each candidate;
- score candidates by whether, and where, the known relevant passage appears in the ranking (for example, within the Top-10).

Form preference pairs so that candidates producing stronger retrieval are the preferred responses. State your preference criterion explicitly (e.g., rank of the relevant passage, or reciprocal rank) and specify how you handle queries where all or none of the candidates succeed. Train with DPO and evaluate the resulting model on TREC DL 2019.

5. Evaluation Protocol

Report all three configurations (zero-shot, SFT, SFT + DPO) on TREC DL 2019 using **MAP**, **nDCG@10**, and **Recall@1000**, with rankings computed to depth 1000, using the reference commands in Appendix A. Present the results in a single comparison table and discuss the differences across configurations.

6. Deliverables

1. **Public GitHub repository** containing the source code, documentation, and end-to-end reproduction instructions (training and evaluation).
2. **Technical report** covering methodology, model selection, implementation details, hyperparameter choices, sample-size justification, experimental setup, quantitative results, analysis, limitations, and future work.

7. Assessment Criteria and Use of AI Tools

Submissions are assessed on: correctness, code quality, reproducibility, experimental methodology, quality of analysis, and justification of design decisions. As part of the subsequent interview stage, technical questions may cover your code, methodology, implementation details, and experimental findings.

AI coding assistants may be used during development. You are, however, expected to fully understand every component of your implementation, as these aspects may be discussed in detail during the subsequent interview stage.

Appendix A. Reference Commands (Pyserini)

These commands reproduce the retrieval and evaluation pipeline on TREC DL 2019.

Query construction. The expanded query written to your topics file follows Query2Doc:

```
reformulated_query = (original_query × 5) + LLM_generated_pseudo_passage
```

Write these to a tab-separated topics file, one query per line: `<qid>\t<reformulated_query>`.

Generate a run on the TREC DL 2019 queries. After you produce your reformulated topics file, replace `--topics dl19-passages` with the path to that file (for example `--topics reformulated_queries.tsv`). The baseline command below uses the built-in DL 2019 topics as a starting point:

```
python -m pyserini.search.lucene \
  --threads 16 --batch-size 128 \
  --index msmarco-v1-passage \
  --topics dl19-passages \
  --output run.msmarco-v1-passage.bm25-default.dl19.txt \
  --bm25 --k1 0.9 --b 0.4
```

Evaluate the run. Here `dl19-passages` is the qrels reference and stays unchanged. MAP and Recall pass `-l 2` to binarize the graded judgments at relevance grade ≥ 2 ; `nDCG@10` uses the graded judgments directly:

```
python -m pyserini.eval.trec_eval -c -l 2 -m map dl19-passages \
  run.msmarco-v1-passage.bm25-default.dl19.txt
python -m pyserini.eval.trec_eval -c -m ndcg_cut.10 dl19-passages \
  run.msmarco-v1-passage.bm25-default.dl19.txt
python -m pyserini.eval.trec_eval -c -l 2 -m recall.1000 dl19-passages \
  run.msmarco-v1-passage.bm25-default.dl19.txt
```

MS MARCO Passage Collection

First, agree to terms and conditions

Terms and Conditions

The MS MARCO datasets are intended for non-commercial research purposes only to promote advancement in the field of artificial intelligence and related areas, and is made available free of charge without extending any license or other intellectual property rights. The dataset is provided "as is" without warranty and usage of the data has risks since we may not own the underlying rights in the documents. We are not be liable for any damages related to use of the dataset. Feedback is voluntarily given and can be used as we see fit. Upon violation of any of these terms, your rights to use the dataset will end automatically.

If you encounter any difficulties with accessing the dataset, please do not hesitate to inform us at ms-marco@microsoft.com. We are committed to ensuring that you have seamless access to all necessary resources and will promptly address any issues you may experience. Please contact us if you own any of the documents made available but do not want them in this dataset. We will remove the data accordingly. If you have questions about use of the dataset or any research outputs in your products or services, we encourage you to undertake your own independent legal review. For other questions, please feel free to contact us.

☒ I agree to terms and conditions. Upon accepting links to dataset will become available.

Submit

Passage Retrieval:RETIRED(10/26/2018-01/01/2023)

Based on the passages and questions in the [Question Answering Dataset](#), a passage ranking task was formulated. There are 8.8 million passages and the goal is to rank based on their relevance.

Relevance labels are derived from what passages was marked as having the answer in the QnA dataset making this one of the largest relevance datasets ever.

This dataset is the focus of the [2020 and 2019 TREC Deep Learning Track](#) and has been used as a teaching aid for [ACM SIGIR/SIGKDD AFIRM Summer School on Machine Learning for Data Mining and Search](#).

In 2020 we release a set of cleaned and formatted clicks for all documents in the collection. This collection of 20 million clicks is called ORCAS.

Tasks

1. Passage Re-Ranking: Given a candidate top 1000 passages as retrieved by BM25, re-rank passage by relevance.
2. Passage Full Ranking: Given a corpus of 8.8m passages generate a candidate top 1000 passages sorted by relevance.

Relevant Links

[NIST Judgments for TREC 2019 Deep Learning Track](#)[Overview of the TREC 2019 deep learning track](#)[Paper](#)[ORCAS Dataset](#)[MSMARCO Passage Ranking Github](#)[TREC 2020 Deep Learning](#)[TREC 2019 Deep Learning](#)

Dataset Download links

[Collection\(10/26/2018\)](#)[Full Web Documents\(05/02/2019\)](#)[Queries\(11/13/2018\)](#)[Qrels Train\(11/28/2018\)](#)[Qrels Dev\(11/28/2018\)](#)

You would need the following files to be downloaded for training:

File	Fields (tab-separated, in order)
Collection	passage_id, passage_text
Queries	qid, query_text
Qrels	qid, 0, doc_id, label

In Qrels, the second column (0) is the unused TREC iteration field; doc_id is the passage_id from the collection; and every label is 1, meaning that passage is judged relevant for the given qid. Use the training queries and training qrels for fine-tuning (Stages 2 and 3).